

Statistics & Linear Algebra — Theory Answers



Statistics & Linear Algebra — Theory Short Questions



Goal: Neeche answers report-style me likhe gaye hain — simple Hinglish, proper math formulas, examples, emoji, aur mini-graphs ke saath.

1. Mean, Median, and Mode 📌

Mean ka matlab average hota hai: saare values add karke total number of values se divide karte hain. **Median** sorted data ka middle value hota hai, aur **Mode** wo value hoti hai jo sabse zyada baar repeat hoti hai. Real life example: agar students ke marks `50, 60, 60, 70, 90` hain, to mean overall performance batata hai, median middle performance batata hai, aur mode most common marks batata hai.

Formula:

$$\text{Mean} = \bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

Example:

Data: 50, 60, 60, 70, 90

Mean = $(50 + 60 + 60 + 70 + 90) / 5 = 66$

Median = 60

Mode = 60

Mini graph:

Marks frequency 

50		■	
60		■	← Mode
70		■	
90		■	

2. Variance vs Standard Deviation

Variance batata hai ki data values mean se kitni spread out hain, lekin iska unit squared hota hai. **Standard Deviation** variance ka square root hota hai, isliye iska unit original data jaisa hi hota hai aur interpret karna easy hota hai. Simple words me: variance spread ka mathematical measure hai, aur standard deviation practical spread batata hai.

Population variance formula:

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Population standard deviation formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

Sample variance formula:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Intuition graph:

Low spread:

10 11 12 13 14

↑ mean ke paas

High spread:

2 8 12 18 25
↑ values zyada scattered



Standard deviation zyada ho to data mean se zyada door-door spread hota hai.

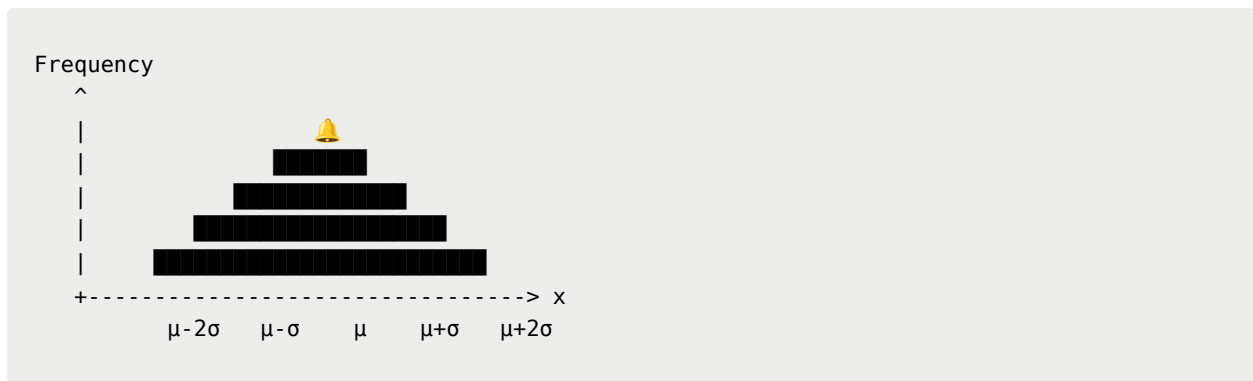
3. Normal Distribution 🔔

Normal Distribution ek bell-shaped distribution hota hai jisme most values mean ke aas-paas hoti hain, aur extreme values kam hoti hain. Isme mean, median, aur mode approximately same hote hain. Practical use case: human height, exam scores, measurement errors, aur machine learning me noise modeling normal distribution follow kar sakte hain.

Formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Bell curve graph:



68–95–99.7 rule:

$\mu \pm 1\sigma \rightarrow$ about 68%
 $\mu \pm 2\sigma \rightarrow$ about 95%

$\mu \pm 3\sigma \rightarrow$ about 99.7%

4. Skewness and Kurtosis

Skewness batata hai ki distribution left ya right side me tilted hai ya nahi. Agar tail right side lambi ho to **positive skew**, aur left side lambi ho to **negative skew**.

Kurtosis batata hai ki distribution ka peak kitna sharp/flat hai aur tails kitni heavy hain.

Skewness formula:

$$\text{Skewness} = \frac{E[(X - \mu)^3]}{\sigma^3}$$

Kurtosis formula:

$$\text{Kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4}$$

Skewness intuition:

Positive skew $\rightarrow$



Mean > Median > Mode

Negative skew $\rightarrow$



Mean < Median < Mode

Kurtosis intuition:

High kurtosis: sharp peak + heavy tails



Low kurtosis: flatter curve



5. Probability: Empirical vs Theoretical Probability 🎲

Probability kisi event ke hone ki chance ko measure karta hai, usually 0 se 1 ke beech. **Theoretical probability** formula se calculate hoti hai jab outcomes equally likely ho, jaise fair coin toss. **Empirical probability** actual experiments ya observations ke data se nikalti hai.

Theoretical probability formula:

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

Empirical probability formula:

$$P(E) = \frac{\text{Number of times event occurred}}{\text{Total number of trials}}$$

Example:

Coin toss 🎲

Theoretical:

$$P(\text{Head}) = 1/2 = 0.5$$

Empirical:

100 tosses me 56 heads aaye

$$P(\text{Head}) = 56/100 = 0.56$$

Mini graph:

Probability scale:

0 ----- 0.5 ----- 1
Impossible Even chance Certain

6. Independent vs Dependent Events

Independent events me ek event ka result dusre event ko affect nahi karta.

Example: coin toss me pehle toss ka result second toss ko affect nahi karta.

Dependent events me ek event ka result dusre event ki probability change kar deta hai, jaise bag se bina replacement ke balls nikalna.

Independent events formula:

$$P(A \cap B) = P(A) \times P(B)$$

Dependent events formula:

$$P(A \cap B) = P(A) \times P(B|A)$$

Examples:

Independent:

Coin Toss 1 = Head 🟡

Coin Toss 2 = Head 🟡

$$P(H \text{ and } H) = 1/2 \times 1/2 = 1/4$$

Dependent:

Bag me 3 red, 2 blue balls 🍒

$$P(\text{First red}) = 3/5$$

Agar red remove ho gayi:

$$P(\text{Second red}) = 2/4$$

Visual:

Independent:

A happens ✅ → B probability same

Dependent:

A happens ✅ → B probability changes 🔄

7. Intuition of Bayes Theorem

Bayes Theorem hume batata hai ki new evidence milne ke baad kisi event ki probability kaise update hoti hai. Daily life example: agar kisi person ko fever hai, to pehle hum sochte hain cold ho sakta hai, lekin test result positive aaye to disease ki probability update ho jaati hai. Simple words me, Bayes theorem old belief + new evidence = updated belief.

Formula:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Where:

P(A) = prior probability
P(B|A) = likelihood
P(A|B) = posterior probability
P(B) = evidence probability

Daily life example:

Before test:
Disease chance = low 🙄

After positive test:
Disease chance increases 📈

Bayes:
Old belief + Evidence = Updated belief

Flow graph:

Prior belief 🧠
+
New evidence 🔍
↓
Updated probability 📊

8. Eigenvalue and Eigenvector 🧭

Eigenvector ek special vector hota hai jo linear transformation ke baad direction change nahi karta, sirf stretch ya shrink hota hai. **Eigenvalue** batata hai ki eigenvector kitna stretch ya shrink hua. Simple example: agar transformation ke baad vector same direction me double ho jaye, to eigenvalue **2** hai.

Main formula:

$$A\vec{v} = \lambda\vec{v}$$

Where:

```
A = matrix / linear transformation  
v = eigenvector  
λ = eigenvalue
```

Visual intuition:

```
Normal vector:  
Before: ↗  
After: → direction changed ❌  
  
Eigenvector:  
Before: ↗  
After: ↗ same direction, length changed ✅
```

Example matrix:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Eigenvector:

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Calculation:

$$A\vec{v} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

So, eigenvalue:

$$\lambda = 2$$



Eigenvectors and eigenvalues PCA, machine learning, Google PageRank, image compression, and physics me useful hote hain.



Quick Summary Table

Topic	Simple Meaning	Key Formula
Mean	Average value	$\bar{x} = \frac{\sum x_i}{n}$
Median	Middle value	Sort data first
Mode	Most repeated value	Highest frequency
Variance	Squared spread	$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$
Std. Deviation	Practical spread	$\sigma = \sqrt{\sigma^2}$
Normal Distribution	Bell-shaped data	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Skewness	Tilt of distribution	$\frac{E[(X-\mu)^3]}{\sigma^3}$
Kurtosis	Peak/tail heaviness	$\frac{E[(X-\mu)^4]}{\sigma^4}$
Probability	Chance of event	$P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$
Bayes Theorem	Updated probability	$P(A B) = \frac{P(B A)P(A)}{P(B)}$
Eigenvector	Direction stays same	$A\vec{v} = \lambda\vec{v}$
Eigenvalue	Stretch/shrink factor	λ



Report tip: Har answer me definition + formula + real-life example likhne se answer strong aur complete dikhta hai.